

What is a Determinant

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Abstract

We derive the determinant of an $n \times n$ matrix from three axioms motivated by oriented volume, prove $|A| = |A^T|$ by reindexing the permutation sum, then compute the derivative of the determinant (*Jacobi's formula*) and apply it to deduce *Liouville's theorem* on the volume evolution of linear flows $\dot{\mathbf{x}} = A\mathbf{x}$.

1 From oriented volume to the determinant formula

The determinant of an $n \times n$ matrix A is the *signed* factor by which the volume of the unit cube is changed upon the action of A . (We will see that the sign is forced by the algebra; for now treat it as a feature, not a bug.) The cube is transformed into a parallelepiped. The n edges of the unit cube meeting at the origin are the standard basis vectors: $\{[1, 0, \dots, 0], [0, 1, 0, \dots, 0], \dots, [0, \dots, 0, 1]\}$. When A acts on these vectors through matrix multiplication it produces the column vectors of A – by the definition of matrix multiplication.

So, to find the determinant – the stretching factor of A – we need to compute the volume of the parallelepiped formed from the column vectors of A . This is enough as the stretching factor is the ratio of the change in volume after applying A to the volume of what we started with. Since the starting point is the unit cube which has a volume of 1, the ratio *is* the volume of the stretched unit cube under the action of A .

Given n vectors in R^n , how do we find a formula for the parallelepiped volume? A good way is to use the properties of the function to help find a formula.

Let's call our volume function $V(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. Here's some properties we should agree on:

Multi-linear Property: V is linear in each slot. That is, it is a multi-linear function.

Degenerate Property: If any two edges are the same then the volume is 0.

Volume of Canonical Unit Cube: The volume of the unit cube using standard ordering of unit edges, $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$, is 1.

Let's go through each property. We are not deriving these from a prior definition of geometric volume; rather, we are *postulating* a function V with these properties, motivated by what oriented volume should look like.

Scaling is intuitive: if we scale any slot (edge) by a factor k then the volume of the resulting parallelepiped should scale by k . Additivity in each slot is more subtle and is genuinely a postulate rather than a visible fact about parallelepipeds — but it is the assumption that lets the rest of the theory go through, and the resulting function turns out to coincide with geometric volume in the cases ($n = 2, 3$) where we can check.

The degeneracy property is clear, if we have duplicate edges then a parallelogram in R^2 collapses to a line segment and consequently the area is zero. Similarly for higher dimensions.

The third property is clearly true, but why the need to care about the ordering of the standard edges? We show why by looking at a property implied by the multi-linearity and degeneracy properties.

Let's focus on R^3 as the general case is similar. Notice that

$$V(\mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_3) = 0 \quad (1)$$

because if the same vector is in two slots the volume is 0. But by the multi-linearity (1) becomes:

$$\begin{aligned} 0 &= V(\mathbf{x}_1, \mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_3) + V(\mathbf{x}_2, \mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_3) \\ &= V(\mathbf{x}_1, \mathbf{x}_1, \mathbf{x}_3) + V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) + V(\mathbf{x}_2, \mathbf{x}_1, \mathbf{x}_3) + V(\mathbf{x}_2, \mathbf{x}_2, \mathbf{x}_3) \\ &= V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) + V(\mathbf{x}_2, \mathbf{x}_1, \mathbf{x}_3) \end{aligned} \quad (2)$$

From this we see that

$$V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = -V(\mathbf{x}_2, \mathbf{x}_1, \mathbf{x}_3) \quad (3)$$

That is, V is *anti-symmetric*.

So, the reason the third property requires the unit cube to be ordered is that order matters because of this anti-symmetry property. *That is, a consequence of our assumptions is that the volume function will be signed!*

Let's now use these two properties to derive a formula for the volume function.

Each of the vectors: $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ can be written in terms of an orthonormal basis: $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ as

$$\begin{aligned} \mathbf{x}_1 &= a_{1,1}\mathbf{e}_1 + a_{2,1}\mathbf{e}_2 + a_{3,1}\mathbf{e}_3 \\ \mathbf{x}_2 &= a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2 + a_{3,2}\mathbf{e}_3 \\ \mathbf{x}_3 &= a_{1,3}\mathbf{e}_1 + a_{2,3}\mathbf{e}_2 + a_{3,3}\mathbf{e}_3 \end{aligned}$$

If $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are the mapping of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ by matrix A , then A is related to the $\{a_{i,j}\}_{i,j=1}^n$ by:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \quad (4)$$

So, coming up with a formula for the volume function in terms of the $\{a_{i,j}\}_{i,j=1}^n$ will translate directly to a formula for the determinant of A .

To illustrate the process of finding the volume we will do it for the case when $n = 2$. In this case we have:

$$\begin{aligned}
V(\mathbf{x}_1, \mathbf{x}_2) &= V(a_{1,1}\mathbf{e}_1 + a_{2,1}\mathbf{e}_2, a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2) \\
&= V(a_{1,1}\mathbf{e}_1, a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2) + V(a_{2,1}\mathbf{e}_2, a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2) \\
&= V(a_{1,1}\mathbf{e}_1, a_{1,2}\mathbf{e}_1) + V(a_{1,1}\mathbf{e}_1, a_{2,2}\mathbf{e}_2) \\
&\quad + V(a_{2,1}\mathbf{e}_2, a_{1,2}\mathbf{e}_1) + V(a_{2,1}\mathbf{e}_2, a_{2,2}\mathbf{e}_2) \\
&= a_{1,1}a_{1,2}V(\mathbf{e}_1, \mathbf{e}_1) + a_{1,1}a_{2,2}V(\mathbf{e}_1, \mathbf{e}_2) \\
&\quad + a_{2,1}a_{1,2}V(\mathbf{e}_2, \mathbf{e}_1) + a_{2,1}a_{2,2}V(\mathbf{e}_2, \mathbf{e}_2)
\end{aligned}$$

Notice that we could pull out $\{a_{i,j}\}_{i,j=1}^n$ because of multi-linearity – linearity refers not just to addition, but also scalar multiplication. Also, we have shown that the degeneracy condition implies anti-symmetry: $V(\mathbf{e}_1, \mathbf{e}_2) = -V(\mathbf{e}_2, \mathbf{e}_1)$. Finally, we know the volume (area in this case) of $V(\mathbf{e}_1, \mathbf{e}_2)$. It represents the volume of the unit cube (unit square in our case) and its volume is 1. Therefore,

$$V(\mathbf{x}_1, \mathbf{x}_2) = a_{1,1}a_{2,2} - a_{1,2}a_{2,1} \quad (5)$$

This corresponds with the usual definition of the determinant of the matrix A .

In the general case, we will have to do all of the permutations of the indices of a upon re-ordering the base vectors into standard form.

Let's see what this looks like in R^3 .

$$\begin{aligned}
V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) &= V(a_{1,1}\mathbf{e}_1 + a_{2,1}\mathbf{e}_2 + a_{3,1}\mathbf{e}_3, \\
&\quad a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2 + a_{3,2}\mathbf{e}_3, \\
&\quad a_{1,3}\mathbf{e}_1 + a_{2,3}\mathbf{e}_2 + a_{3,3}\mathbf{e}_3)
\end{aligned} \quad (6)$$

By multi-linearity this can be decomposed into the following:

$$V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 a_{i,1}a_{j,2}a_{k,3}V(\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k) \quad (7)$$

There are 3^3 terms to consider. In the general case, there will be n^n terms to deal with – this is not looking promising. But notice, not all terms, $V(\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k)$ survive. Any such term with a duplicate $\mathbf{e}_i$ will have value 0 because of the degeneracy condition. What terms do survive? The terms where the indices, i, j, k , differ. That is, i, j, k are *permutations* of 1, 2, 3.¹ In the general case this reduces the number of terms in the sum from n^n to $n!$. In the general case, the sum becomes:²

$$V(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) = \sum_{\sigma \in S_n} a_{\sigma(1),1}a_{\sigma(2),2} \dots a_{\sigma(n),n}V(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)}) \quad (8)$$

Now, how do we compute each of the terms: $V(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)})$? We know that we can switch the values of any two of the n slots and the value of V will change signs. It turns out that for any permutation,

¹More generally, permutations on n objects are one to one mappings from the set of integers $\{1, 2, \dots, n\}$ to itself.

² S_n is the symmetric group of n objects. It consists of all permutations that can be performed on n objects.

one can apply some minimal number of these switches (called *transpositions*) to get back to the canonical ordering $V(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$. So the value of $V(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)})$ should be the value of $V(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ times $(-1)^k$, where k is the number of minimal transpositions needed to get to the canonical ordering of indices. We denote the sign of σ by $\text{sgn}(\sigma) \in \{-1, +1\}$:³ And since we know the volume of the canonical cube is 1, the formula for the parallelepiped – and hence the determinant of the matrix A – becomes:

$$\begin{aligned}
|A| &= V(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) \\
&= V(a_{1,1}\mathbf{e}_1 + a_{2,1}\mathbf{e}_2 + \dots + a_{n,1}\mathbf{e}_n, \\
&\quad a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2 + \dots + a_{n,2}\mathbf{e}_n, \\
&\quad \vdots \\
&\quad a_{1,n}\mathbf{e}_1 + a_{2,n}\mathbf{e}_2 + \dots + a_{n,n}\mathbf{e}_n) \quad (\text{expanding vectors into components}) \\
&= \sum_{j_1=1}^n \sum_{j_2=1}^n \dots \sum_{j_n=1}^n a_{j_1,1} a_{j_2,2} \dots a_{j_n,n} V(\mathbf{e}_{j_1}, \mathbf{e}_{j_2}, \dots, \mathbf{e}_{j_n}) \quad (\text{by the Multi-linearity Property}) \\
&= \sum_{\sigma \in S_n} a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} V(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)}) \quad (\text{by the Degeneracy Property}) \\
&= \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} V(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) \quad (\text{after placing edges in Canonical Order}) \\
&= \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} \quad (\text{Volume of Canonical Unit Cube is 1}) \tag{9}
\end{aligned}$$

Note. There is a bug with this process. From our reasoning, it follows that the volume of $V(\mathbf{e}_2, \mathbf{e}_1)$ is -1. It seems like this ordering of edges is as valid a way to describe the unit square, so why is the area negative? We can fix this “bug” by claiming that it is, in fact, a feature. We can claim that we haven’t just computed the area/volume, we have computed the *oriented volume*. If you want the volume you need only take the absolute value of the oriented volume. Mathematicians call this a *feature* rather than a bug; claiming that the determinant gives you more than just the volume, you get the *oriented volume*.

We write the final formula down for reference but now with the caveat that the determinant of a matrix is not a positive stretching factor – the amount that the volume of the unit cube is stretched under A , but a *signed* stretching factor. Again, to find the magnitude of the stretching, take the absolute value of the determinant.

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} \tag{10}$$

³The mapping $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is a group homomorphism; $\text{sgn}(\sigma) = -1$ if k is odd and $\text{sgn}(\sigma) = +1$ if k is even, where k is the number of transpositions in any factorization of σ (this parity is well-defined).

1.1 Determinant of A^T

Notice that the formula (10) can also be written as:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(T(\sigma)) a_{T(\sigma)(1),1} a_{T(\sigma)(2),2} \cdots a_{T(\sigma)(n),n} \quad (11)$$

Here, T is any one to one mapping from S_n to itself. Why? Because the sum in (10) is the sum over all elements of S_n . The mapping T just rearranges that sum. One such choice for T is the mapping that sends σ to its inverse, σ^{-1} . In this case, (11) becomes:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma^{-1}) a_{\sigma^{-1}(1),1} a_{\sigma^{-1}(2),2} \cdots a_{\sigma^{-1}(n),n} \quad (12)$$

Reorder the factors in each product (multiplication is commutative) and re-index by $i = \sigma^{-1}(k)$, so that $k = \sigma(i)$:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma^{-1}) a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} \quad (13)$$

Letting $a'_{i,j}$ denote the (i,j) entry of A^T (so $a'_{i,j} = a_{j,i}$), we have $a_{i,\sigma(i)} = a'_{\sigma(i),i}$, hence:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma^{-1}) a'_{\sigma(1),1} a'_{\sigma(2),2} \cdots a'_{\sigma(n),n} \quad (14)$$

Since $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is a group homomorphism, $\text{sgn}(\sigma) \cdot \text{sgn}(\sigma^{-1}) = \text{sgn}(\epsilon) = 1$,⁴ so $\text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma)$ (since ± 1 are self-inverse). With this, (14) may be written:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a'_{\sigma(1),1} a'_{\sigma(2),2} \cdots a'_{\sigma(n),n} \quad (15)$$

But this is, by definition, nothing more than the determinant of A^T . From this we conclude that $|A| = |A^T|$.

2 Derivative of the Determinant

The determinant is a function from the space of matrices to R ; that is, $\det : R^{n \times n} \rightarrow R$. Therefore, its derivative at a particular “point” A , L_A , should satisfy:⁵

$$|A + H| = |A| + L_A(H) + o(H) \quad (16)$$

Here $\lim_{H \rightarrow 0} \frac{|o(H)|}{\|H\|} = 0$, where $\|H\|$ is any matrix norm and L_A is a linear mapping from $R^{n \times n}$ to R .

⁴ ϵ denotes the identity in S_n .

⁵Recall that for a function $f : R \rightarrow R$, the derivative at a point x is a linear function, $Df_x : R \rightarrow R$, such that $f(x+h) = f(x) + Df_x(h) + o(h)$.

To start, let's try to find the derivative for an especially simple “point”, the identity matrix, I .

$$|I + H| = \prod_{i=1}^n (1 + H_{i,i}) + o(H) \quad (17)$$

Only the identity permutation contributes to the linear-in- H part of $|I + H|$. To see this, take any non-identity $\sigma \in S_n$. Its non-fixed points form cycles of length ≥ 2 , so there are at least two indices i with $\sigma(i) \neq i$. For each such i , the entry $(I + H)_{\sigma(i),i} = H_{\sigma(i),i}$ is off-diagonal (no 1 contribution). Hence the product for σ contains at least two off-diagonal H entries, so it is $O(\|H\|^2) = o(\|H\|)$. This product can be further decomposed into constant, linear-in- H , and $o(H)$ terms:

$$\begin{aligned} |I + H| &= 1 + \sum_{i=1}^n H_{i,i} + o(H) \\ &= |I| + \text{tr}(H) + o(H) \end{aligned} \quad (18)$$

So, the linear function acting on H to produce a number is the trace operator.

Now, let's do the same computations for a more general “point”, an $n \times n$ matrix, A , with the stipulation that A is invertible.

$$\begin{aligned} |A + H| &= |A(I + A^{-1}H)| \\ &= |A| |I + A^{-1}H| \quad (\text{Multiplicative Property: } |AB| = |A||B|) \\ &= |A| (|I| + \text{tr}(A^{-1}H) + o(H)) \quad (\text{From (18)}) \\ &= |A| + |A|\text{tr}(A^{-1}H) + o(H) \end{aligned} \quad (19)$$

Therefore, the derivative of the determinant at a non-singular “point”, A , is the linear function $L_A : R^{n \times n} \rightarrow R$ defined by:

$$L_A(B) = |A|\text{tr}(A^{-1}B) \quad (20)$$

This is known as *Jacobi's formula* for the derivative of the determinant.⁶

Exercise. Check that L_A is a linear function.

⁶The invertibility assumption on A can be removed. The general form of Jacobi's formula is $L_A(B) = \text{tr}(\text{adj}(A)B)$, where $\text{adj}(A)$ is the adjugate (transpose of the cofactor matrix). This expression is polynomial in the entries of A and so is defined for all A , singular or not. When A is invertible, $\text{adj}(A) = |A|A^{-1}$ and the two forms agree; when A is singular, the adjugate form is the right one.

3 Applications: Liouville's Theorem

Suppose that a dynamical system is deforming n vectors in R^n over time by the following linear *vector differential equations*:

$$\frac{d\mathbf{x}_i(t)}{dt} = A\mathbf{x}_i(t) \quad i \in [1, n] \quad (21)$$

$$\mathbf{x}_i(0) = \mathbf{x}_{0,i} \quad i \in [1, n] \quad (22)$$

Question. How is the volume of the parallelepiped formed from the vectors $\{\mathbf{x}_{0,i}\}_{i=1}^n$ evolving in time?

To start, we could collect these vector equations into one linear *matrix differential equation*:

$$\frac{dX(t)}{dt} = AX(t) \quad (23)$$

$$X(0) = X_0 \quad (24)$$

Here, the n vectors form the columns of the matrix $X(0)$. The columns of $X(t)$ represent the evolution of the vectors, $\{\mathbf{x}_i\}_{i=1}^n$, in time. The determinant of $X(t)$, $|X(t)|$, represents the evolution of the volume of the parallelepiped represented by these vectors.

To answer the question posed, we need to track $|X(t)|$. To do this, we find a differential equation that $|X(t)|$ satisfies and then solve.

$$\frac{d|X(t)|}{dt} = |X(t)| \operatorname{tr} \left(X(t)^{-1} \frac{dX(t)}{dt} \right) \quad (\text{by (20) and the chain rule})$$

$$\frac{d|X(t)|}{dt} = |X(t)| \operatorname{tr} (X(t)^{-1} AX(t)) \quad (\text{by (23)})$$

$$\frac{d|X(t)|}{dt} = |X(t)| \operatorname{tr}(A) \quad (\text{by the property of trace: } \operatorname{tr}(ABC) = \operatorname{tr}(CAB))$$

Therefore, $|X(t)|$ satisfies the following linear *scalar differential equation*:

$$\frac{d|X(t)|}{dt} = \operatorname{tr}(A) |X(t)| \quad (25)$$

$$|X(0)| = |X_0| \quad (26)$$

This is much easier to solve, the solution is $|X(t)| = |X_0| e^{\operatorname{tr}(A)t}$. This is *Liouville's formula*; it is a special case of Liouville's theorem on phase-space volume preservation in dynamical systems. The chain-rule step above used $X(t)^{-1}$; this is justified post hoc, since $|X(t)| = |X_0| e^{\operatorname{tr}(A)t}$ is non-zero whenever $|X_0| \neq 0$, so $X(t)$ remains invertible along the flow.

Question. What kind of systems (defined by A) leave the volumes unchanged?

From the differential equation $\frac{d|X(t)|}{dt} = \operatorname{tr}(A) |X(t)|$ and its solution $|X(t)| = |X_0| e^{\operatorname{tr}(A)t}$, the volume is preserved exactly when $\operatorname{tr}(A) = 0$. An important sub-class is anti-symmetric matrices: when $A^T = -A$, then $a_{ii} = -a_{ii}$ forces $a_{ii} = 0$ for every i , so $\operatorname{tr}(A) = 0$ and volumes are preserved.

Anti-symmetric A in fact gives a *stronger* conclusion than mere volume preservation. From $A^T = -A$ we have $\langle A\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{x}, A^T \mathbf{x} \rangle = -\langle \mathbf{x}, A\mathbf{x} \rangle = -\langle A\mathbf{x}, \mathbf{x} \rangle$, so $\langle A\mathbf{x}, \mathbf{x} \rangle = 0$. Applied to the trajectory:

$$\left\langle \mathbf{x}(t), \frac{d\mathbf{x}(t)}{dt} \right\rangle = \langle \mathbf{x}(t), A\mathbf{x}(t) \rangle = 0, \quad (27)$$

which gives $\frac{d}{dt} \|\mathbf{x}(t)\|^2 = 2 \langle \mathbf{x}, \dot{\mathbf{x}} \rangle = 0$: the *individual length* of each evolving vector is preserved, not just the volume of the parallelepiped they span.

Exercise. In this case, what kind of matrix is $X(t)$?